

HAMZA KADDOUR

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Summary

Geospatial Machine Learning Engineer with 4 years of professional and research experience building predictive models, data-intensive applications, and reproducible AI workflows. Current work applies deep learning to satellite and geospatial data for environmental monitoring, integrating SAR, optical imagery, and auxiliary variables across end-to-end pipelines. Experienced in Python, PyTorch, AWS, model evaluation, explainability, and translating technical research into operational outputs for multidisciplinary teams

Skills

- **Programming Languages:** Python, JavaScript, C++, SQL, NoSQL, MySQL, AWS, Linux Administration, Windows Server
- **Tools:** PyTorch, TensorFlow, Docker, Git, Jira, MySQL, NumPy, Pandas, Splunk, Tableau, AWS
- **Concepts:** Machine Learning, Data Visualization, Deep Learning, Experimental Methods, Predictive Modeling, Transformers, Data Pre-processing, Reinforcement Learning, Semantic Segmentation, Statistical Methods, Cloud Security, CI/CD, Software Architecture, Database, Cybersecurity, DevSecOps, Model Evaluation

Languages

- English(Full professional proficiency (TOEFL iBT))
- French(Native)
- Arabic(Native)

Experience

Stevens Institute of Technology

Jul 2025 - Present

Machine Learning Engineer

Hoboken, NJ

- Develop end-to-end geospatial AI workflows for river-ice monitoring using Sentinel-1 SAR, Sentinel-2 optical imagery, VIIRS, NISAR, and environmental variables, producing aligned training datasets and operational model outputs.
- Design and evaluate deep-learning segmentation models including U-Net/UNet++, Swin Transformer, and temporal transformer architectures; conduct ablation, robustness, and explainability analyses using IoU, F1, SHAP, Grad-CAM, and channel occlusion.
- Engineer reproducible Python/GDAL/rasterio/AWS pipelines for ingestion, reprojection, multi-sensor alignment, training, inference, and reporting; automated processing reduced manual effort by more than 70%.
- Automated IT tasks including deployment, VM network monitoring, and database management (DynamoDB and MySQL) using Python, JavaScript, and YAML, resulting in improved task efficiency.
- Evaluated and tested code security through unit-testing and systematic code revisions to ensure robust and secure software development.
- Maintained network, IT infrastructure, and high performance computing systems to ensure reliable, scalable operations.

Idaho State University

Jan 2024 - Jun 2025

Data Scientist

Idaho Falls, ID

- Developed a Web Security Scoring application using Python, JavaScript, HTML, CSS, and ReactJS, enhancing security assessments for 1000 domains monthly and integrating data pre-processing and visualization practices.
- Engineered a hierarchical face recognition and deepfake image classification system using machine learning, applying predictive modeling and experimental methods to achieve 91.6% precision and strengthen model evaluation.
- Delivered programming courses that reinforced object-oriented programming (OOP) and software engineering fundamentals, strengthening theoretical foundations related to data processing and ML model building
- Explored reinforcement learning applications to enhance cybersecurity and Quality of Service in 5G architectures using Deep Q Networks with Optuna, achieving a 0.12 average reward and demonstrating algorithm optimization practices.
- Constructed Java-based simulations complemented by MySQL data management to accurately predict network performance and inform data-driven decision-making.

Responsible Cyber

Feb 2022 - Dec 2023

Cybersecurity Software Engineer

Singapore

- Integrated third-party tools, security APIs, and internal scripts (using Linux Bash and PowerShell) into a web application to scan over 10,000 domains, ensuring data integrity and streamlined update processes.
- Revamped legacy code bases and implemented modern development standards; reduced memory consumption by 98% and accelerated database operations by 50x using MySQL and data manipulation libraries.
- Developed and tested automation scripts in Java and Python to enhance security analytics, showcasing skills in data pre-processing and model-based threat detection.
- Conducted comprehensive unit-testing on new AWS Lambda functions and step functions with Python prior to deployment to ensure robust performance.
- Constructed Python-based predictive models for security compliance evaluations with 87.1% accuracy and 87.9% F1-Score, leveraging techniques in machine learning algorithm selection and model optimization.

2SB

Sep 2021 - Jan 2022

Cloud Engineer Intern

France

- Simulated cloud-based services (NextGen Proxy & CASB) using Netskope, contributing to enhanced service testing and validation through data-driven insights.

- Monitored and managed network incidents to safeguard company and client networks using TCP/IP, DNS, Firewall, and Splunk, effectively integrating data pre-processing and visualization for improved incident response.
- ITSERV

DevOps Engineer Intern

Jun 2021 - Sep 2021

Tunis, Tunisia

 - Integrated agile security tests (SAST, DAST, and IAST) into a CI/CD pipeline using GitLab, maintaining a 98.5% pipeline speed and demonstrating effective algorithmic evaluation of code security.
 - Assisted over 20 clients in troubleshooting cloud security issues using Python and Splunk, enhancing overall security posture and compliance through systematic data analysis.

Education

- Idaho State University

Master of Science, Computer Engineering

Jan 2024 - May 2025

Pocatello, ID

 - **GPA:** 4/4
 - **Coursework:** Software Engineering, C/C++, Object-Oriented-Programming, JAVA, Docker, Data Science, REST API, Advanced Algorithms, Complexity, Cybersecurity, Cryptography, Python, Deep Learning, Reinforcement Learning

- Higher University of Communication of Tunis (SUP COM)

Bachelor of Science, Computer and Telecommunication Engineering

Sep 2017 - Sep 2022

Tunis, Tunisia

 - **GPA:** 4/4
 - **Coursework:** Data Structures and Algorithms, Software Development, Python, C, SQL, JAVA, Operating Systems (Linux, Windows), Virtualization, Spring Boot, Secure Coding, Network protocols, Data Science, Machine Learning, Internet of Things, Software Engineering, English

Publications

- H. Kaddour, B. Cherif, M. Temimi."Deep Learning-Based Segmentation of Sentinel-1 SAR Satellite Imagery for River Ice Detection with Weak Optical Supervision".*IEEE International Geoscience and Remote Sensing Symposium (IGARSS), 2026, Washington D.C. USA* Accepted, presented, and to be published
- H. Kaddour et al.."Generalizable Deep Reinforcement Learning-Based Intelligent Handover in Indoor WiGig Networks".*2025 IEEE 102nd Vehicular Technology Conference (VTC2025-Fall), Chengdu, China* DOI: 10.1109/VTC2025-Fall65116.2025.11310213
- H. Kaddour et al."Evaluating the performance of machine learning-based classification models for IoT intrusion detection".*2024 IEEE Opportunity Research Scholars Symposium (ORSS), Atlanta, GA, USA* DOI: 10.1109/ORSS62274.2024.10697949
- H. Kaddour, I.G. Olaveson, C.J. Krome, M.M Fouda."Altering 5G network parameters using deep reinforcement learning to optimize qos and security".*2024 IEEE Virtual Conference on Communications (VCC), NY, USA* DOI: 10.1109/VCC63113.2024.10914411
- H. Kaddour, M.I. Ibrahim, Z.M. Fadlullah, M.M Fouda."Multi-Domain Deep Reinforcement Learning for Cyber-Physical Systems Security on the WDT Testbed Dataset".*2025 3rd International Conference on Artificial Intelligence, Blockchain, and Internet of Things (AIBThings), Mt Pleasant, MI, USA* DOI: 10.1109/AIBThings66987.2025.11296162